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Natalia A. Agapova

ML Engineer

Education

Innopolis University — Innopolis, Russia — 2022 – 2026

- **Major:** BSc in Applied Artificial Intelligence
- **Programming Coursework:** Operating Systems, Distributed and Network Programming, Information Retrieval, Industrial C++
- **AI/ML Coursework:** Practical Machine Learning and Deep Learning, Introduction to Computer Vision, Explainable, Interpretable and Fair AI, Generative Artificial Intelligence, Natural Language Processing

Employment

Croissan Studio — **Jun 2025 to present**

Client Work, NeuroArtist — **Frontend Developer** — *May 2026 to present*

- Rebuilding the front-end for Croissan Studio's main website redesign in stages as designers finish each section, restructuring the UX along the way and increasing clicks on the "Contact us" button by 13%.
- Shipping new features and contributing to front-end client work.
- **Leveraged knowledge** in React, Next.js, and TypeScript to rebuild the site's front-end, styled with Tailwind and built out with Radix UI components and Lucide icons.

NeuroArtist, Pocket NeuroArtist — **ML Software Engineer** — *Aug 2025 to May 2026*

- Worked as the ML engineer on client product teams, including NeuroArtist and Pocket NeuroArtist, reworking NeuroArtist's core model and fully fine-tuning transformer language models for client products.
- Improved generated-image quality for Pocket NeuroArtist through iterative testing, growing its user numbers by about 5.5%.
- **Leveraged knowledge** in Python and PyTorch, fine-tuning Hugging Face Transformers and building generative-AI image models with Diffusers, using LoRA and prompt engineering, and prototyping in Figma.

NeuroArtist — **Software Engineer Intern** — *Jun 2025 to Aug 2025*

- Worked on the NeuroArtist website and contributed to client projects, for example a tourism website and the Moscow 2030 event.
- Helped with ML tasks on the team – including tracking down a bug in the image upscaler that had been mistaken for a model issue – and was offered an ML Engineer role after the internship.
- **Leveraged knowledge** in Python for ML debugging and JavaScript/React for the website, versioned with Git and GitHub.

- *Letter of recommendation from the studio's CEO* →

Asimov Lab (startup) — **Frontend Team Lead & ML Engineer** — **Feb to Sep 2024**

- Led the frontend team to build the MVP website end to end for an AI platform that generates tests for teachers and students, and was one of six ML developers behind its core model.
- **Leveraged knowledge** in React and TypeScript for the frontend, and Python with scikit-learn for the model.

Freelance — **ML Engineer** — **2022 to present**

- Applied ML work for independent clients, including Russian Post, Innopolis City Administration, Innopolis University, Lyceum Innopolis, and others.
- **Leveraged knowledge** in Python, PyTorch, and scikit-learn for client ML projects.

Personal Projects

Fair Resume Screening — NLP · Explainability · Debiasing · Published, first author

- Built a BERT resume classifier (9 IT categories, HeadHunter data) to audit and reduce bias in automated screening; found city-swapping flipped 7.7% of decisions (60.9% baseline) – published this analysis as first author.
- **Utilized:** Python, PyTorch, BERT, Captum, pandas.

Emotion Detection — Computer Vision · On-device Deployment

- Trained a CNN for facial emotion recognition (58.7% accuracy, 60.8% with ensembling), shrunk to 1.15 ms on-device (Core ML, ONNX Runtime).
- **Utilized:** Python, PyTorch, Core ML, ONNX Runtime.

Personal Knowledge System — RAG · NLP · Embeddings

- Built a local RAG assistant that cites exact file/page from your own PDFs, powered by Llama 3.2 – nothing leaves your machine.
- **Utilized:** Python, SentenceTransformer, ChromaDB, Ollama.

Tend — On-device ML · Chrome Extension

- Chrome extension that groups tabs on-device via keyword rules and a confidence-gated MiniLM model, plus an opt-in Claude mode.
- **Utilized:** JavaScript, transformers.js (MiniLM), Claude API.

Publications

Agapova, N. A.; Lukmanov, R. A. "Improving Fairness in AI-Powered Recruitment: An Interpretable Resume Screening System", 2026. doi:10.66693/mathai.1017

Kholodov, Y. A.; Salloum, H.; Agapova, N. A. "Quantum Annealing Approaches to Breaking RSA Encryption", *Voprosy kiberbezopasnosti*, 2025. doi:10.21681/2311-3456-2025-4-127-133

Skills

ML / AI: *(proficient)*: Python, PyTorch, BERT, Hugging Face Transformers, Diffusers, LoRA fine-tuning, generative AI, LLMs (Ollama), Captum, model evaluation, RAG (ChromaDB, SentenceTransformer) *(familiar)*: scikit-learn, pandas, Core ML, ONNX Runtime
Languages & Frameworks: *(proficient)*: React *(familiar)*: TypeScript, Next.js, Tailwind
Tools: *(proficient)*: Figma, Tilda, Git/GitHub *(familiar)*: Radix UI, Lucide
Languages: Russian C2 (native) · English C2 · Ukrainian B1 · French A2 · Tatar A2 · Korean A1